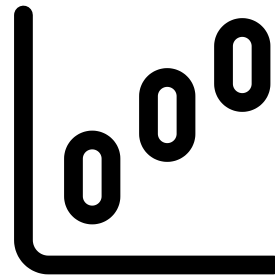


Basic Statistics

Basic Statistics

- Deep learning requires understanding basic Statistics
 - Introduction of notation
 - Review of basic concepts



Probability: An Example

- Toss a coin. Will it land on heads or tails?

We cannot perfectly observe the world e.g. weight of coin, hand would touch the surface of coin, force. That is the reason coin toss looks random. Hence 50-50 chance.

Note: Side on top has higher chance.



Basic Notation

$$P(X = a)$$

Random variable X taking specific value “ a ” the outcome is called event where P is probability

Basic Notation: An Example

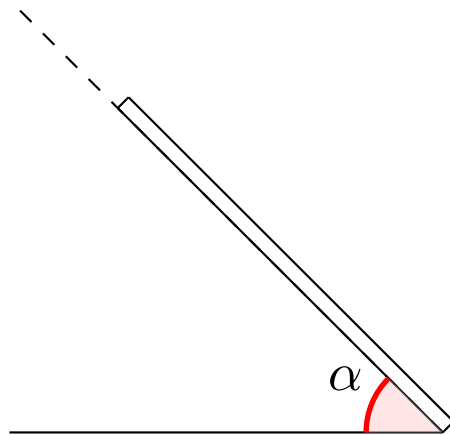
$$P(\overbrace{X = a}^{\text{an event}})$$

a random variable a specific value



Probability Density

- What is $P(Y = \alpha)$?
 - not defined
- Cumulative probability: $P(\alpha_1 \leq Y < \alpha_2)$
- Probability density: $p(Y = \alpha)$
 - $$p(Y = \alpha) = \frac{P(\alpha - \epsilon \leq Y < \alpha + \epsilon)}{2\epsilon}$$



Unified Notation

Symbol	Description
$P(x) = P(X = x)$	Probability of a discrete event
$P(x) = p(X = x)$	Probability density of a continuous event

We will not make distinction between actual probability of discrete event vs continuous event

Properties of $P(x)$

A function $P(x)$ that captures the probabilities of any value x .

Property

Non-negativity

$$0 \leq P(x)$$

Boundedness (discrete only)

$$0 \leq P(x) \leq 1$$

Summation (discrete)

$$E_P[1] = \sum_x P(x) = 1$$

Summation (continuous)

$$E_P[1] = \int P(x)dx = 1$$

Probability Distributions

$$P(x) = P(\overbrace{X = x}^{\text{an event}})$$

a random variable a specific value

- What is P ?
 - A function
 - Discrete $P : \{c_1, c_2, \dots, c_n\} \rightarrow [0, 1]$
 - Continuous $P : \mathbb{R} \rightarrow \mathbb{R}$
 - Called a **probability distribution**



Examples of Probability Distributions: Visual World



Examples of Probability Distributions: Weather

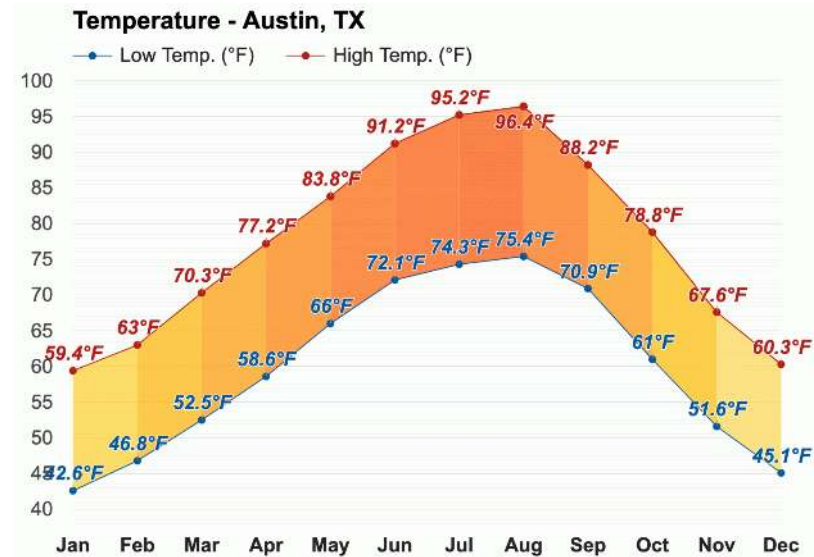


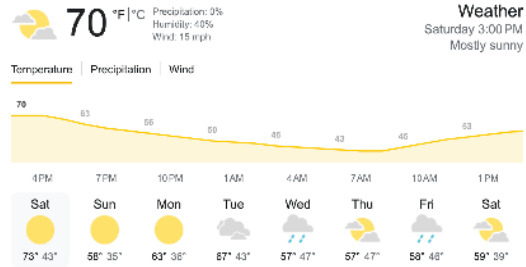
Image source: Weather U.S.

Three Types of Distribution

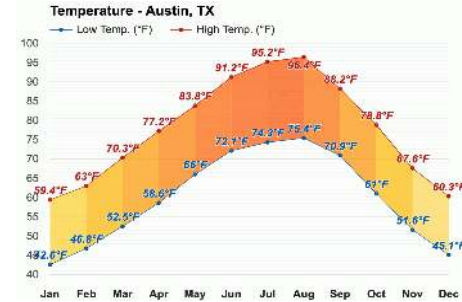
Data Generating



Empirical



Model



Basic Statistic Concepts

Expectation

$$E_{x \sim P} [f(x)] = \begin{cases} \sum_x P(x) f(x) & \text{discrete } P \\ \int_x P(x) f(x) dx & \text{continuous } P \end{cases}$$

Expectation of function $f(x)$ under distribution P is in the discrete case is weight sum of discrete events. i.e. multiply probability with function at that event. For continuous its integral.

Expectation - Short Forms

- $E_P [f(x)]$
- $E [f(x)]$
- $E_P [f]$
- $E [f]$

Linearity of Expectation

$$E[f(x) + g(x)] = E[f(x)] + E[g(x)]$$

$$E[\alpha f(x)] = \alpha E[f(x)]$$

Mean and Variance

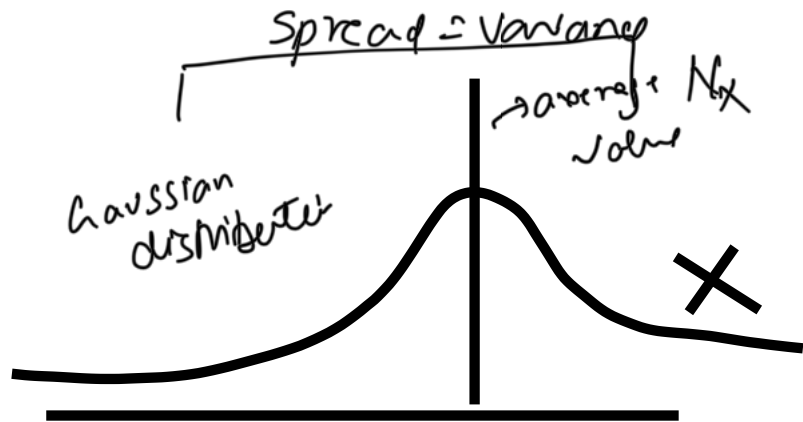
Mean:

Average value a certain distribution takes

$$\mu_x = E_{x \sim P}[x]$$

Variance:

$$\sigma_x^2 = \text{Var}_{x \sim P}[x] = E_{x \sim P}[(x - \mu_x)^2]$$



N_x - height of the bell curve at any specific point x .

probability of hitting one exact number is zero $x = ?$

Unified Notations: A Word of Warning

Discrete distribution	Continuous distribution	
$Var_{x \sim P} [x]$ is always finite	$Var_{x \sim P} [x]$ can be infinite	
$P(X)$ always less than 1	$P(X)$ can be larger than 1	This is probability density
...	...	

Sampling

$$x \sim P$$



Example of Sampling



Example of Sampling



Example of Sampling

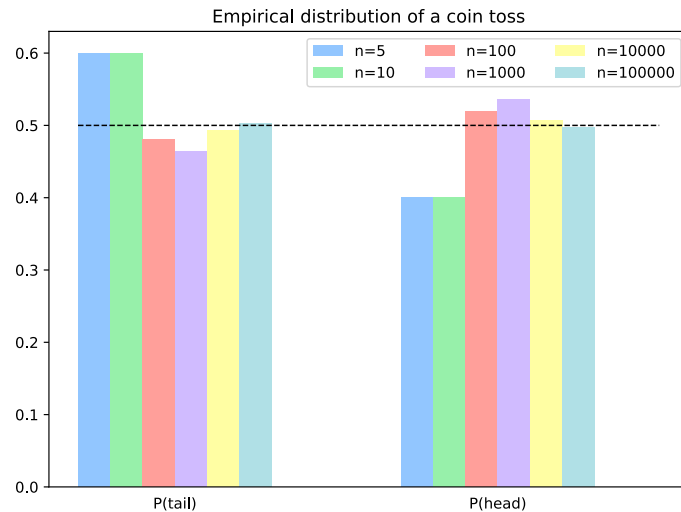


Bias in Samples

- Samples are always biased

- For infinite samples:

empirical distribution = data generating distribution

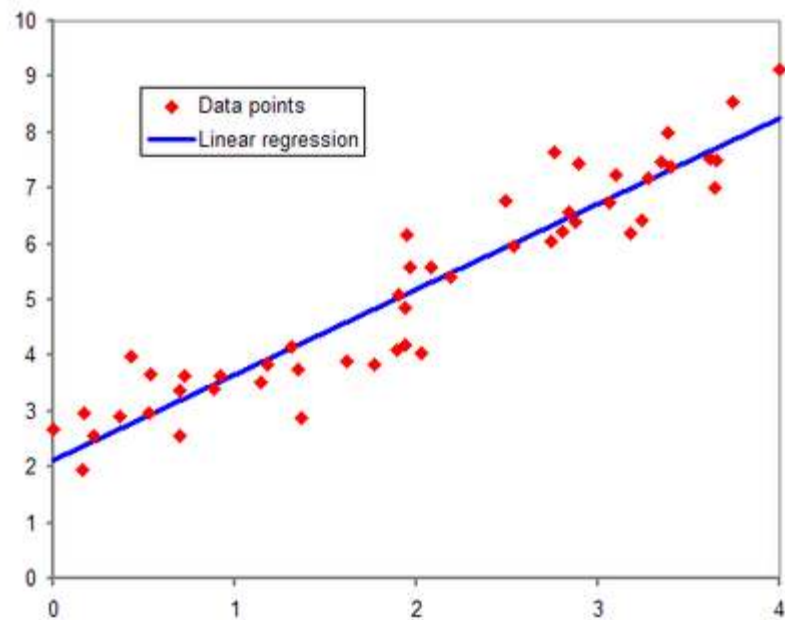


Statistical Models

Regression Model

$$f_{\theta} : \mathbb{R}^n \rightarrow \mathbb{R}^d$$

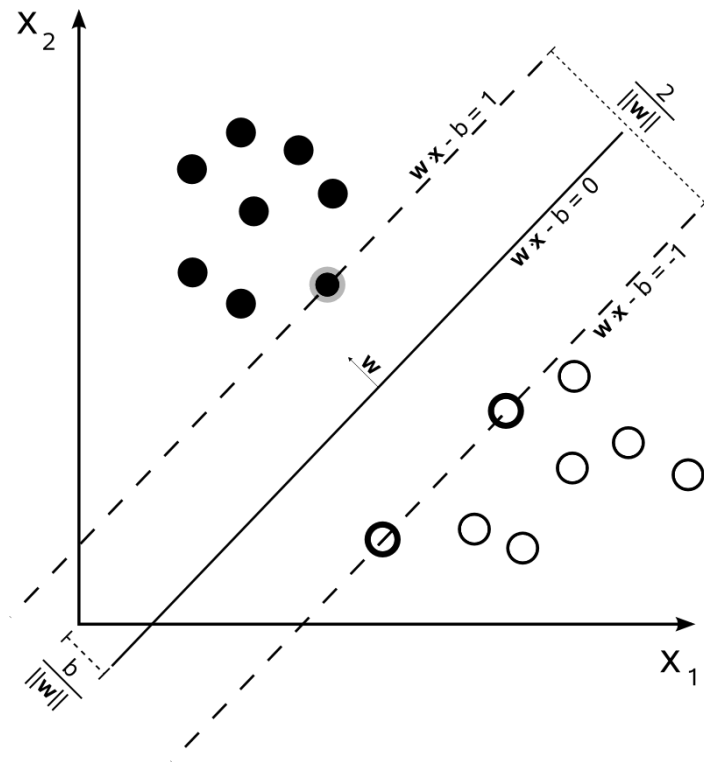
Function that map certain input to output. Here its parameterized by theta that map real input to real output.



Classification Model

$$f_{\theta} : \mathbb{R}^n \rightarrow P(X), P(X) \subset \mathbb{R}^d$$

Map real value to probability to class output or probability over possible classes or discrete event.



Statistical Model Summary

$$f_{\theta} : X \rightarrow Y$$

- X : input space
- Y : output space
- θ : model parameters

In machine learning:

- Goal: find the optimal **parameter** θ
- How: learn from **data**

Data

Unlabeled Data:

$$D = \{x_1, x_2, \dots\} \quad \text{where } x_i \sim P_D.$$

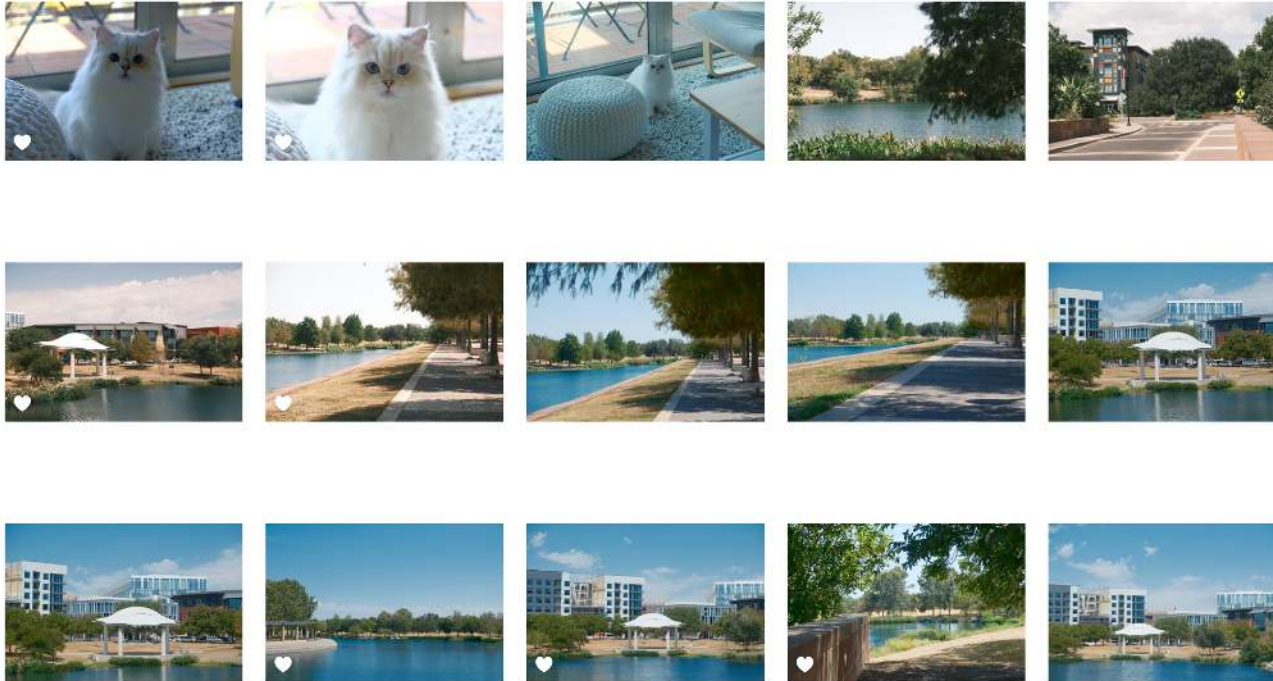
Labeled Data:

$$D = \{(x_1, l_1), (x_2, l_2), \dots\} \quad \text{where } (x_i, l_i) \sim P_D,$$

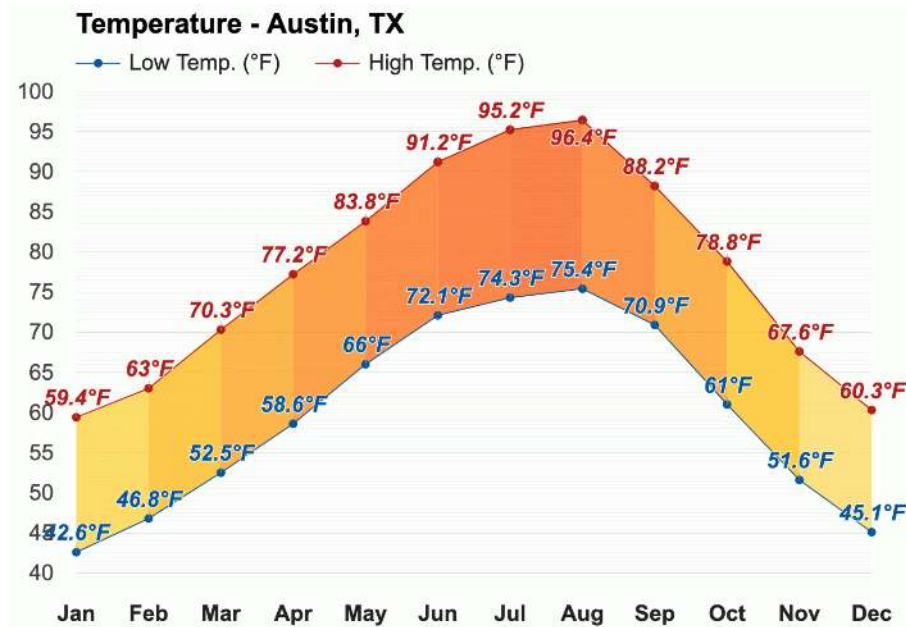
Examples of Data: Internet

A screenshot of the Wikipedia article "Web page". The top navigation bar includes the Wikipedia logo, a search box, and links for "Create account" and "Log in". Below the header, there's a sidebar on the left with links like "Contents [hide]", "Elements", "Navigation", "Deployment", "See also", and "References". The main content area has a title "Web page" followed by tabs for "Article" and "Talk". A summary line reads "From Wikipedia, the free encyclopedia". The first paragraph defines a web page as a hypertext document on the World Wide Web, delivered by a web server to a user via a web browser. It mentions that a website consists of many such pages linked together under a common domain name. To the right of the text is a small inset image showing a screenshot of a web page about Wikipedia. Below the main text, there are sections for "Elements", "Navigation", and "Deployment", each starting with a brief overview and ending with a reference to further information or related concepts.

Examples of Data: Personal Photo Collection



Examples of Data: Weather



Source: Weather U.S.

Basic Statistics - TL;DR

Model: a function of the data to predict values $f_\theta : X \rightarrow Y$

Expectation: measures the value of f weighted by probability P .

Variance: measures deviation from the mean